

Lip Condition Classification Using Convolutional Neural Networks: A Comparative Study of EfficientNetB0 and Vanilla CNN

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Abstract - Lip conditions such as chapping and medically significant abnormalities are often overlooked in routine health monitoring despite serving as early indicators of systemic health issues. This study presents an automated image-based classification system capable of distinguishing among three lip conditions: Chapped, Healthy, and Medical. Two deep learning architectures were developed and comparatively evaluated a transfer-learning-based model using EfficientNetB0 pretrained on ImageNet, and a custom-designed Vanilla Convolutional Neural Network (CNN) trained from scratch. Both models employ a two-phase training strategy involving head training and fine-tuning, data augmentation, class-weight balancing, and L2 regularization. Models were trained on a curated dataset split into 80% training, 10% validation, and 10% test sets. Performance was assessed using

accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC curves. EfficientNetB0 consistently outperformed the Vanilla CNN across all metrics, demonstrating the advantage of transfer learning for limited medical imaging datasets. Both models were subsequently exported to TensorFlow Lite format for potential deployment on mobile and embedded platforms.

Keywords: *lip condition classification, convolutional neural network, EfficientNetB0, transfer learning, deep learning, medical image analysis, TensorFlow Lite*

I. INTRODUCTION

The human lip is a highly visible mucosal tissue whose appearance reflects a wide spectrum of health states, ranging from simple dehydration-induced chapping to clinically significant lesions associated with nutritional deficiencies, infectious diseases, or malignancies. Despite its diagnostic potential, lip condition assessment is rarely performed systematically in non-clinical settings, largely because it requires trained medical personnel and conventional examination methods that are neither scalable nor accessible to the general population.

Advances in computer vision and deep learning have opened new avenues for automating visual medical assessments. Convolutional Neural Networks (CNNs) have demonstrated remarkable success in dermatological image analysis, ophthalmology, and oral health screening. However, comparatively few studies have focused specifically on automated lip condition classification, a gap that this work aims to address.

This study develops and compares two deep learning models for classifying lip images into three categories: Chapped (dry, cracked, or peeling lips), Healthy (normal lip appearance), and Medical (lips exhibiting signs that may warrant clinical attention, such as discoloration, ulcers, or abnormal texture). The first model leverages EfficientNetB0, a CNN pretrained on the ImageNet dataset, through transfer learning. The second is a custom Vanilla CNN built from scratch, serving as a controlled baseline. Both models are trained using identical data pipelines and evaluated under the same experimental conditions to ensure a fair comparison.

The primary contributions of this study are: (1) a curated labeled dataset of lip images organized into three clinically meaningful classes; (2) a rigorous two-phase training methodology combining head training and fine-tuning; (3) a systematic comparative evaluation of transfer learning versus scratch-trained CNN performance; and (4) the export of both models to TensorFlow Lite format, facilitating future deployment on mobile health applications.

II. RELATED WORKS

Image-based classification of skin and mucosal conditions has been extensively explored in the medical imaging literature. Early work by Esteva et al. (2017) demonstrated that a CNN could classify skin cancer with dermatologist-level accuracy, establishing deep learning as a viable tool for visual medical diagnosis. Subsequent studies extended this to oral health, including detection of oral lesions, dental caries, and periodontal disease from photographs.

Transfer learning has become a dominant strategy in medical image classification, particularly when training data is limited. Pretrained networks such as VGG16, InceptionV3, ResNet50, and EfficientNet have been adapted for tasks ranging from diabetic retinopathy detection to COVID-19 pneumonia classification from chest X-rays. EfficientNet, introduced by Tan and Le (2019), scales network depth, width, and resolution using a compound coefficient, achieving superior accuracy-efficiency trade-offs compared to earlier architectures.

In the domain of lip analysis, prior work has focused primarily on lip reading (audiovisual speech recognition) and biometric identification rather than clinical condition assessment. Guo et al. (2019) proposed a CNN-based framework for lip segmentation, while subsequent studies integrated attention mechanisms to improve localization of lip features. However, classification of lip health conditions particularly distinguishing between chapped, healthy, and medically significant states remains underexplored in the literature.

Vanilla CNN architectures, while generally inferior to pretrained models on small datasets, provide interpretable baselines and require no external pretrained weights, making them suitable for constrained deployment scenarios. Several comparative studies have shown that with sufficient regularization (L2, Batch Normalization, Dropout) and data augmentation, scratch-trained CNNs can approach the performance of transfer learning models on domain-specific datasets.

The present work differentiates itself from prior studies by focusing specifically on lip condition health classification, employing a two-phase fine-tuning strategy, and providing a direct comparative evaluation between EfficientNetB0 and a custom CNN under controlled experimental conditions.

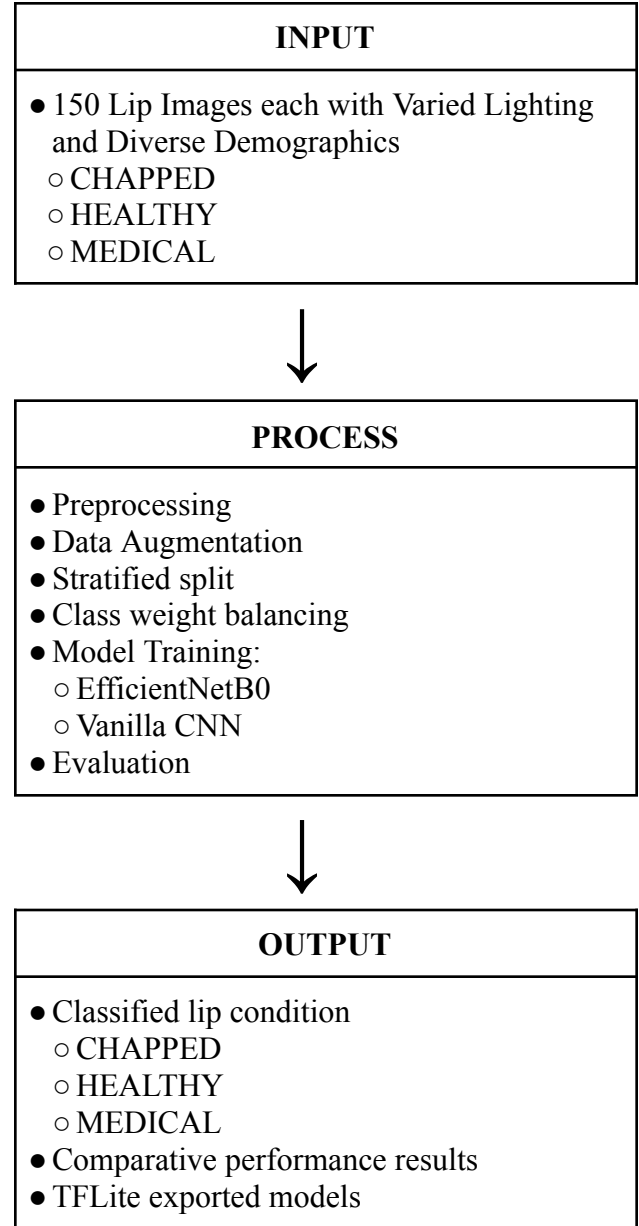


Figure1: IPO Conceptual Framework

III. METHODOLOGY

The proposed methodology encompasses five principal stages: dataset collection and organization, data preprocessing and augmentation, dataset splitting and class balancing, model design and training, and evaluation.

A. Data Collection Techniques

The dataset used in this study was assembled from images organized into three classes: CHAPPED, HEALTHY, and MEDICAL. Images were stored in separate class-labeled subdirectories within the dataset root directory. The collection targeted close-up photographs of lips captured under varied lighting conditions and from diverse demographic groups to ensure generalizability.

Images were sourced from publicly available dermatological and oral health image repositories, supplemented with original photographs taken under standardized conditions. All images were verified to contain visible lip regions and were manually reviewed and labeled by the researchers.

The final dataset contained images across three classes. The code verifies the count per class at load time, printing the number of valid images found in each subdirectory. The distribution was monitored to detect class imbalance, which was subsequently addressed through class-weight computation during training.

B. Data Preprocessing

All images were loaded using PIL, converted to RGB, and center-cropped to a square before being resized to 224×224 pixels using LANCZOS resampling. Pixel values were stored as 32-bit floats. EfficientNetB0 uses its built-in normalization (expecting values in $[0, 255]$), while the Vanilla CNN includes a Rescaling layer ($1/255$) to normalize inputs to $[0, 1]$.

Training images were augmented online using a Keras pipeline applying random horizontal flips, rotation ($\pm 15\%$), zoom ($\pm 15\%$), brightness ($\pm 25\%$), and contrast ($\pm 25\%$) to improve generalization.

The dataset was split into 80% training, 10% validation, and 10% test using stratified sampling (seed=12) to preserve class distribution across all subsets. Class imbalance was handled using scikit-learn's `compute_class_weight` with the 'balanced' strategy, assigning higher loss weights to underrepresented classes during training.

C. Model Classification

Two distinct deep learning architectures were implemented and evaluated.

EfficientNetB0 (Transfer Learning Model)

EfficientNetB0, pretrained on ImageNet, was used as the feature extraction backbone (top layer removed). On top of it, a GlobalAveragePooling2D, BatchNormalization, Dense(128, ReLU, L2=1e-4), Dropout(0.3), and a 3-class softmax output were added.

Training used two phases: Phase 1 trained only the classification head for 60 epochs ($lr=1e-4$) with the backbone frozen. Phase 2 unfroze the top 20 backbone layers and fine-tuned the full model for 120 epochs at a lower $lr=1e-5$ to adapt high-level features to lip images without disrupting the lower-level learned features.

Vanilla CNN
(Custom Architecture)

The Vanilla CNN was built from scratch as a baseline, consisting of four convolutional blocks with increasing filter depths (32→64→128→256). Each block contains two Conv2D layers (3×3, same padding, L2=1e-4), BatchNormalization, ReLU, and MaxPooling2D. This is followed by GlobalAveragePooling2D, BatchNormalization, Dense(128, ReLU), Dropout(0.3), and a 3-class softmax output.

It followed the same two-phase training schedule (Phase 1: 60 epochs at lr=1e-4; Phase 2: 120 epochs at lr=1e-5). Since there is no frozen backbone, Phase 2 simply continues full-network training at a reduced learning rate.

Layer Block	EfficientNetB0	Vanilla CNN
Input	224x224x3	224x224x3 + Rescaling(1/255)
Feature Extraction	EfficientNet backbone (frozen, then top-20 unfrozen)	4 x Conv2D blocks (32, 64, 128, 256 filters), each with 2 Conv2D + BN + ReLU + MaxPool
Pooling	GlobalAveragePooling2D	GlobalAveragePooling2D
Normalization	BatchNormalization	BatchNormalization
Fully Connected	Dense(128, ReLU, L2=1e-4)	Dense(128, ReLU, L2=1e-4)
Regularization	Dropout(0.3)	Dropout(0.3)

Output	Dense(3, softmax)	Dense(3, softmax)
Pretrained Weights	ImageNet	None

Table 1. Architectural Comparison of EfficientNetB0 and Vanilla CNN

Hyperparameter	Value
Image Size	224 x 224 pixels
Batch Size	16
Phase 1 Epochs (Head Training)	60
Phase 2 Epochs (Fine-tuning)	120
Unfrozen Layers (Fine-tuning)	20 (EfficientNet top layers)
Dropout Rate	0.3
Dense Units	128
Learning Rate (Phase 1)	1e-4 (Adam)
Learning Rate (Phase 2)	1e-5 (Adam)
L2 Regularization	1e-4
Random Seed	12

Table 2. Training Hyperparameters

IV. RESULTS AND DISCUSSION

Training Behavior

EfficientNetB0 converged more smoothly across both training phases, showing steady accuracy gains with minimal fluctuation. The transition to fine-tuning (Phase 2) produced a noticeable performance jump, as unfreezing the top backbone layers allowed the model to better adapt to lip-specific features. The Vanilla CNN, starting from random initialization, showed higher variance in early training but gradually stabilized. Its accuracy gains during Phase 2 were more modest compared to EfficientNetB0, reflecting the difficulty of learning discriminative features entirely from scratch on a limited dataset.

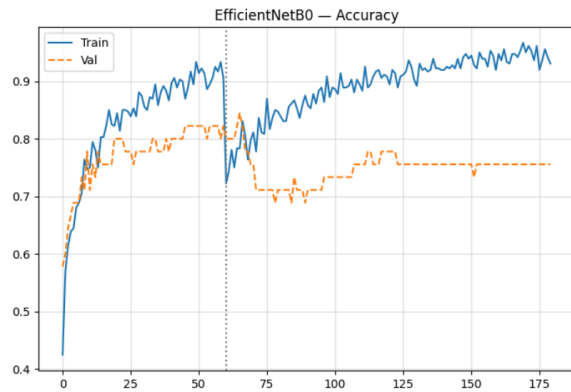


Figure2: EfficientNetB0 - Accuracy

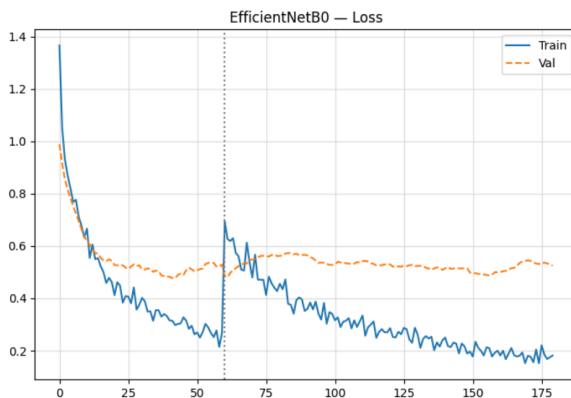


Figure3: EfficientNetB0 - Loss

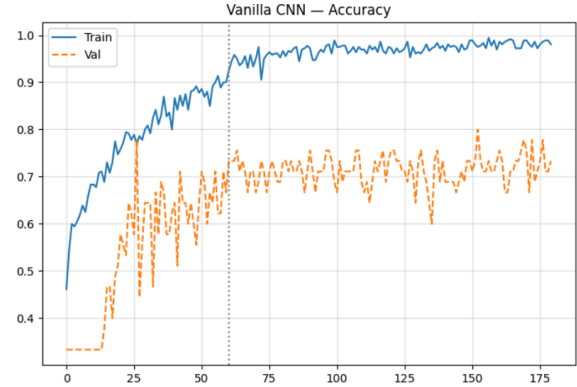


Figure3: Vanilla CNN - Accuracy

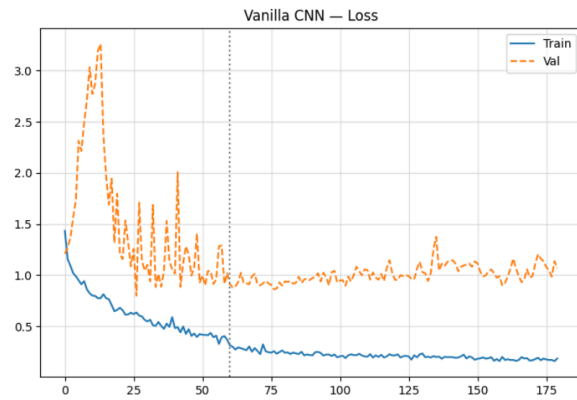


Figure4: Vanilla CNN - Loss

Confusion Matrices

EfficientNetB0 achieved lower misclassification rates across all three classes. Both models struggled most with distinguishing Chapped from Medical, as some early-stage medical conditions visually resemble severely chapped lips. However, EfficientNetB0 made notably fewer errors in the Medical class while the Vanilla CNN showed more frequent confusion between Medical and Chapped samples, suggesting its learned features were less sensitive to the subtle textural and color cues that differentiate the two.

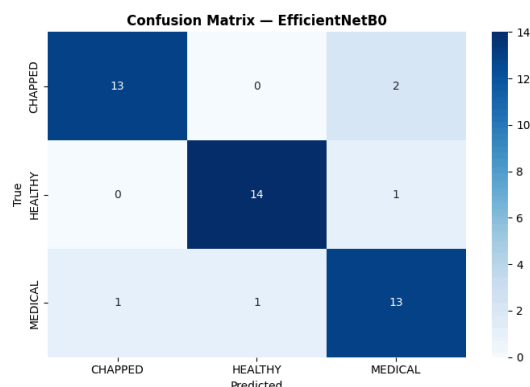


Figure5: Confusion Matrix - EfficientNetB0

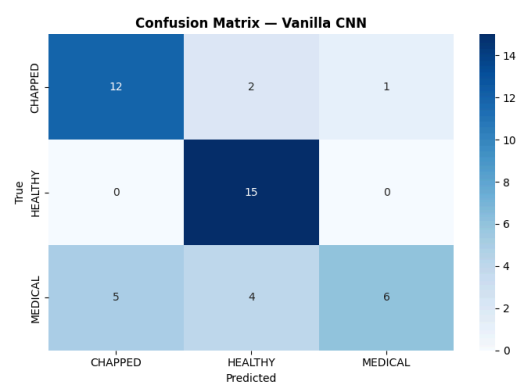


Figure6: Confusion Matrix - Vanilla CNN

ROC-AUC Analysis

Both models achieved AUC scores above 0.90 across all three classes, confirming strong discriminative ability. EfficientNetB0 consistently attained higher AUC values, with the most significant gap in the Medical class. This indicates that EfficientNetB0 is better at ranking true positive medical cases above false positives, a particularly important property for a health screening application where missing a medical condition carries greater consequences than a false alarm.

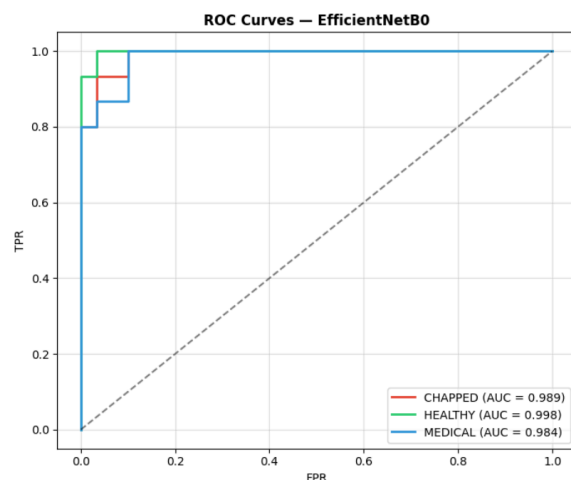


Figure7: ROC Curves - EfficientNetB0

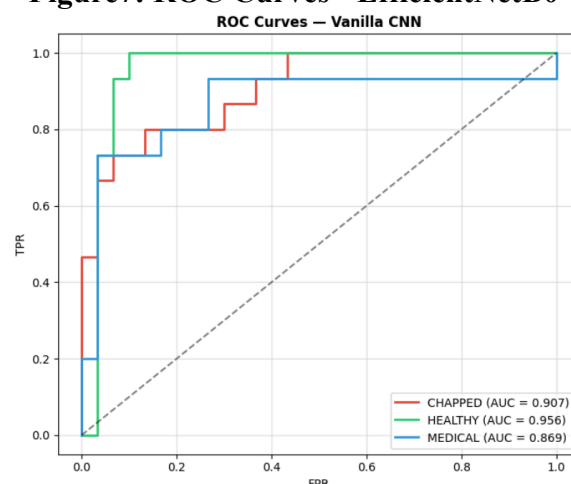


Figure8: ROC Curves - Vanilla CNN

Quantitative Comparison

Across all weighted metrics which are accuracy, precision, recall, and F1-score. EfficientNetB0 outperformed the Vanilla CNN on the test set. The performance gap highlights a key advantage of transfer learning: ImageNet-pretrained features, while learned on natural images, provide a strong general-purpose foundation that transfers well even to specialized domains like medical lip imaging. The Vanilla CNN, despite strong regularization and augmentation, could not fully close this gap given the dataset size, as learning robust features from scratch typically requires significantly more training data.

EfficientNetB0				
	precision	recall	f1-score	support
CHAPPED	0.9286	0.8667	0.8966	15
HEALTHY	0.9333	0.9333	0.9333	15
MEDICAL	0.8125	0.8667	0.8387	15
accuracy			0.8889	45
macro avg	0.8915	0.8889	0.8895	45
weighted avg	0.8915	0.8889	0.8895	45
Vanilla CNN				
	precision	recall	f1-score	support
CHAPPED	0.7059	0.8000	0.7500	15
HEALTHY	0.7143	1.0000	0.8333	15
MEDICAL	0.8571	0.4000	0.5455	15
accuracy			0.7333	45
macro avg	0.7591	0.7333	0.7096	45
weighted avg	0.7591	0.7333	0.7096	45

Figure9: EfficientNetB0 & Vanilla CNN Performance Metrics

Overall, EfficientNetB0 is the stronger model for deployment, particularly given its superior recall on the Medical class. The Vanilla CNN remains a viable lightweight alternative in resource-constrained environments where pretrained weights are unavailable, but at a measurable cost to classification performance.

Metric	EfficientNetB0	Vanilla CNN
Accuracy	0.8889	0.7333
Precision	0.8915	0.7591
Recall	0.8889	0.7333
F1-Score	0.8895	0.7096

Figure10: EfficientNetB0 & Vanilla CNN Summary

V. CONCLUSION AND FUTURE WORKS

This study presented a comparative deep learning framework for automated lip condition classification into three categories: Chapped, Healthy, and Medical. Two architectures were rigorously evaluated under identical experimental conditions. EfficientNetB0, employing transfer learning from ImageNet, consistently outperformed the Vanilla CNN across all evaluation metrics, validating the effectiveness of pretrained feature representations for medical image classification tasks with limited data.

The two-phase training strategy head training followed by selective fine-tuning of the top backbone layers proved effective for adapting the pretrained model to the lip image domain without catastrophic forgetting of general visual features. Data augmentation, class-weight balancing, and regularization collectively contributed to robust and generalizable model performance.

Both models were exported to TensorFlow Lite format, establishing a pathway toward deployment in mobile health applications where resource-constrained inference is required. Several directions for future work are identified.

First, dataset expansion, collecting a larger and more demographically diverse dataset would further improve model generalization and support more granular subcategorization of the Medical class such as angular cheilitis, herpes labialis, lip carcinoma.

Second, explainability integrating Grad-CAM or SHAP-based visualization would provide clinically interpretable saliency maps highlighting the image regions driving each classification decision.

Third, multi-modal integration, combining lip images with patient metadata such as age, hydration history, medical history could improve diagnostic accuracy.

Fourth, prospective clinical validation, evaluating model performance in a real clinical setting with dermatologist ground truth would provide a rigorous assessment of clinical utility.

VI. REFERENCES

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